

Weekly Report 6: Web-Based Real-Time Multiplayer Ludo Game

Date: August 10, 2026

Team Members:

- Shihab Uddin Patwary (ID: 2132685642)
 - Md. Sajidul Islam (ID: 2121935642)
 - Mohiuddin Sefat (ID: 2231193642)
 - Sakiha Zaman (ID: 2233351642)
- Ishraq Hossain Rodro (ID: 2311675042)
- Musanna Ibn Amin (ID: 2233285042)

Executive Summary

This week, we focused heavily on refining the client-side architecture and expanding our deployment options by implementing a serverless, peer-to-peer (WebRTC) room exchange model alongside our core engine. We successfully migrated and polished our frontend UI components, enhanced local pass-and-play features, and streamlined the data channel pipelines for direct browser-to-browser multiplayer sessions without requiring a heavy signaling backend.

Detailed Task Accomplishments

1. Client Architecture & UI/UX Refinement

- **Modular Screen Design:** Finalized distinct view modules for the welcome lobby, user authentication (login/signup), and the main game board interface.
- **Parametric SVG Board Generation:** Programmed a responsive, mathematical layout engine rendering the 15x15 Ludo grid, safe-zone star tiles (\$\star\$), home columns, and color-coded yards dynamically.
- **Interactive Game Controls:** Built real-time UI components including a simulated dice face with authentic pip distributions, animated rolling transitions, dynamic turn-banners, and active player indicator panels.

2. Serverless WebRTC Multiplayer & Room Management

- **P2P Networking Layer:** Integrated a robust RTCPeerConnection data channel layer allowing users to host and join games via manual copy/paste SDP invite and answer code exchanges.
- **Host-Authoritative State Synchronization:** Ensured that game rules are processed authoritatively by the host peer, broadcasting synchronized board states, valid move arrays, and dice outcomes securely to connected joiners.
- **Lobby & Mode Selection:** Configured flexible routing logic supporting both local

multi-user pass-and-play on a single device and remote online sessions.

3. Core Rule Engine & Security Integration

- **Comprehensive Game Rules:** Enforced complete Ludo logic rules including base-exit constraints (requiring a roll of 6), movement across 52 global shared tracks, capture mechanics, and extra turn triggers.
- **Authentication Guards:** Implemented robust client-side session guards via local storage checks (ludo_current_user) to secure private game routing and maintain user profile continuity.

Report compiled and submitted for review by the project team members listed above.